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ORIGINAL PAPER

Spatial variation of earthworm communities and soil organic carbon in temperate agroforestry

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Abstract

The aim of this study was to assess how soil organic C (SOC) stocks and earthworm communities were modified in agroforestry systems compared to treeless control plots and within the agroforestry plots (tree rows vs alleys). We used a network of 13 silvoarable agroforestry sites in France along a north/south gradient. Total earthworm abundance and biomass were significantly higher in the tree rows than those in the control plots, but were not modified in the alleys compared to those in the control plots. Earthworm species richness, Shannon index, and species evenness were significantly higher in the tree rows than those in the alleys. Total abundance of epigeic, epi-anecic, strict anecic, and endogeic was higher in the tree rows. Surprisingly, earthworm individual weight was significantly lower in the tree rows than that in the alleys and in the control plots. SOC stocks were significantly higher in the tree rows compared to that in the control plots across all sites. Despite higher SOC stocks in the tree rows, the amount of available C per earthworm individual was lower compared to those in the control. The absence of disturbance (no tillage, no fertilizers, no pesticides) in the tree rows rather than increased SOC stocks therefore seems to be the main factor explaining the increased total abundance, biomass, and diversity of earthworms. The observed differences in earthworm communities between tree rows and alleys may lead to modified and spatially structured SOC dynamics within agroforestry plots.

Keywords Silvoarable system · Alley cropping · Earthworm abundance · Earthworm diversity · Lumbricidae

Introduction

Earthworms are a major component of the soil fauna in temperate climates (Lavelle 1988). They are usually classified into three main ecological categories having different morphology, physiology, and behavior: epigeic, anecic, and endogeic (Bouché 1972, 1977; Potvin and Lilleskov 2017). Briefly, epigeic earthworms live and feed on surface organic matter; anecic earthworms make vertical or near-vertical burrows into which they incorporate varying amounts of surface organic matter and feed on a mix of surface and soil organic matter; and endogeic earthworms make horizontal or near-horizontal burrows and feed on humified soil organic matter. A distinction between epi-anecic and strict-anecic earthworms are sometimes made (Bouché 1972, 1977). Epi-anecic earthworms burrow few galleries and mainly feed on fresh surface organic matter, whereas strict anecic earthworms mainly feed on soil humified organic matter burrowing a large network of galleries (Bouché and Kretzschmar 1974; Kretzschmar 1977; Ferrière 1980; Jégou et al. 1998, 2001a, b). Depending on their ecological category, earthworms are involved in numerous ecosystem services, such as pedogenesis, soil structure formation, nutrient cycling, and climate regulation through soil organic carbon (SOC) sequestration (Lavelle et al. 2006; Blouin et al. 2013; Bertrand et al. 2015), through their burrowing activity and through their casts that are nutrient-

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rich compared to the bulk soil (Saharan and Singh 1988; Tian et al. 2000; Hmar and Ramanujam 2014).

The assembly of earthworm community is driven by several environmental factors, such as the biogeographical history of the region, broad habitat constraints (microclimate and soil properties), land use constraints, and internal community constraints (competition or facilitation) (Decaëns et al. 2008). In agricultural landscapes, land use and land management can modify earthworm abundance and diversity (Decaëns et al. 2003; Smith et al. 2008b; Pelosi et al. 2009; Cluzeau et al. 2012; Frazão et al. 2017) and consequently impact ecosystem services provided by earthworms (Lavelle 1997, Lavelle et al. 2006; Jouquet et al. 2006). For example, Ponge et al. (2013) observed across 109 sites that agricultural fields exhibited a lower anecic earthworm abundance than grasslands. Slurry application only enhanced endogeic earthworm abundance, whereas epigeic earthworm abundance was not influenced by neither land use nor management. At the plot scale and in cropping systems, Chan (2001) highlighted that anecic abundance tends to decline under tillage, whereas endogeic abundance can increase especially under organic fertilization. A recent meta-analysis confirmed that epigeic and anecic earthworms were the most sensitive ecological groups to conventional tillage (Briones and Schmidt 2017). Several studies have observed that total earthworm abundance, biomass, and richness were greater in field margins than those in cultivated fields (Smith et al. 2008a; Nieminen et al. 2011; Roarty and Schmidt 2013; Crittenden et al. 2015).

Agroforestry systems include very diverse farming systems where trees and crops are grown in intimate combination (Nair 1993). In temperate regions, silvoarable systems associating parallel tree rows and annual intercrops, as well as silvopastoral systems combining trees, pastures, and livestock, are the most widespread types of agroforestry. In temperate silvoarable systems, trees are planted in parallel rows, and the space between trees along the rows is usually untilled, unfertilized, and covered by a natural or sown herbaceous vegetation. In tropical systems, several authors have shown that total earthworm abundance, biomass, and activity were increased in alley cropping compared to those in adjacent agricultural plots (Hauser 1993; Hauser et al. 1998; Fonte et al. 2010). In temperate regions, very few studies have been performed, but Price and Gordon (1999) observed in a silvoarable system in Canada a higher earthworm abundance and biomass in the tree row than those in the cropped alley. Several explanations have been proposed to explain the higher occurrence of earthworms in agroforestry tree rows, especially the shading effect of trees, the lower soil temperature, the higher soil moisture (Tian et al. 2000), the lower level of soil disturbance (Hauser et al. 1998), and the higher amount of food, i.e., organic matter (Araujo and López-Hernández 1999; Frouz et al. 2009). SOC stocks are indeed usually increased in silvoarable systems compared to those in treeless agricultural fields (Chatterjee et al. 2018; de Stefano and Jacobson 2018; Feliciano et al. 2018; Shi et al. 2018; Cardinael et al. 2018a), especially in tree rows (Bambrick et al. 2010; Wotherspoon et al. 2014; Cardinael et al. 2015a, 2017). This additional food supply could positively affect earthworm individual weight (Shipitalo et al. 1988).

Due to the lower level of soil disturbance, the higher amount of food, and buffered microclimatic conditions, tree rows could therefore represent favorable habitats for earthworm communities' development, especially for epigeic and anecic earthworms, potentially enabling them to colonize the cropped alleys. However, previous studies on earthworm communities in agroforestry systems have never taken into account the earthworm ecological categories or species susceptible to respond differently to specific properties in agroforestry system.

The objectives of this study were thus to (i) compare earthworm communities and SOC stocks between agroforestry systems and treeless control plots and to (ii) compare the distribution of earthworm communities and SOC stocks between tree rows and alleys within the agroforestry plots.

Firstly, we hypothesized that total earthworm abundance, biomass, diversity, and earthworm individual weight would be higher in the agroforestry plots compared to those in the control plots, especially in the tree rows. Secondly, we hypothesized that the abundance of epigeic and anecic earthworms would be higher in the tree rows than those in the alleys and control plots.

Material and methods

Site description

We selected a network of 13 agroforestry plots in France along a north/south gradient. Soil properties as well as the soil use and management varied greatly across sites. Briefly, the age of agroforestry sites ranged from 6 to 41 years and tree density from 35 to 200 trees ha^{-1}. In addition, soil pH ranged from 5.8 to 8.4, while clay content ranged from 100 to 530 g kg^{-1}. (Table 1). Each site comprised an agroforestry plot and an adjacent agricultural control plot. Agroforestry alleys and control plots have been managed strictly the same way (soil tillage, crop rotation, fertilization) since the tree planting. A detailed description of the sites can be found in the Supplementary materials. Due to time and budget constraints, not all sites have been sampled for both earthworms and SOC (see below).

Quantification of soil organic carbon stocks

SOC stocks at the CH, ME, RE, SJ, and VZ sites were measured in a previous study using an intensive soil sampling (Cardinael et al. 2017). In this study, we measured SOC stocks

Table 1 Description of the agroforestry sites

Site	Location	Soil texture clay/silt/sand (g kg^{-1})	Soil pH	Age (years)/planting date	Tree row/alley width (m)	Density (trees ha^{-1})	Crop management	Tree species	Crops	Sampling locations	
										C	Earthworms
BE	0° 58′ 37.5″ E, 46° 32′ 24.4″ N	297/325/378	7.1	8/2007	2/26	48	Conventional-reduced tillage	Mixed species	Rapeseed, wheat, barley	AF	AF, C
BO	0° 07′ 45.4″ E, 45° 12′ 39.4″ N	447/346/207	8.1	8/2007	3/27	48	Conventional, tillage	Poplar	Maize	AF, C	AF
CH	1° 17′ 58″ E, 48° 06′ 08″ N	195/705/100	7.0	6/2008	2/24	34	Conventional, tillage	Hybrid walnut	Wheat, rapeseed	AF, C	AF, C
LB	2° 03′ 52.5″ E, 49° 28′ 25.6″ N	246/603/151	8.0	6/2009	2/28	83	No pesticides, reduced tillage	Mixed species	Rapeseed, wheat, barley	AF, C	AF, C
ME	0° 10′ 37″ W, 46° 11′ 54 ″ N	250/645/105	5.8	6/2008	2/27	35	Conventional, tillage	Hybrid walnut	Wheat, rapeseed, sunflower	AF, C	AF, C
PS	0° 12′ 58.9″ W, 47° 49′ 18.1″ N	235/247/518	–	8/2007	3/27	67 or 33	Conventional, no-tillage	Mixed species	Rapeseed, wheat, vetch-peas, barley, sorghum	AF	AF, C
RE	04° 01′ E, 43° 43′ N	175/410/415	8.0	18/1995	2/11	110	Conventional, tillage	Hybrid walnut	Durum wheat, rapeseed, chickpea	AF, C	AF
SA	0° 02′ 53.9″ W, 45° 27′ 06.8″ N	–	–	6/2007	5/20	65	Direct seeding, cover crops	Mixed species	Cereals	–	AF, C
SJ	0° 13′ 57″ W, 46° 00′ 39″ N	530/390/80	7.7	41/1973	2/12	102	Conventional, tillage	Black walnut	Sunflower, wheat, barley	AF, C	AF, C
SJM	0° 02′ 28.7″ W, 46° 54′ 19.0″ N	329/355/316	8.4	9/2006	3/24	35	Conventional, tillage	Mixed species	Maize	AF, C	AF
SM	0° 30′ 42.9″ W, 46° 23′ 57.6″ N	240/612/148	8.0	5 or 7/ 2008 or 2010	2/25	40	Organic, no-tillage	Mixed species	Cereals	–	AF
VER	2° 48′ 56.4″ E, 49° 40′ 03.3″ N	199/684/117	8.1	7/2008	2/28	46	Conventional, reduced tillage	Mixed species	Beetroot, wheat, field bean, rapeseed	AF, C	AF, C
VZ	4° 06′ 37″ E, 44° 03′ 29″ N	100/390/510	8.3	18/1995	2/9	100	Organic, tillage	Hybrid walnut	Rapeseed, wheat, potato, garlic	AF, C	AF

The age corresponds to the age of the agroforestry plot at the time of soil sampling for C analysis, except for the sites with no soil C data where the age corresponds to the time of earthworm sampling. Tree density corresponds to the agroforestry tree density. The crop management was the same in both the alleys and control plots

AF: agroforestry plot, C: control plot (treeless)

BE, Béthines; BO, Bonnes; CH, Châteaudun; LB, Lasalle Beauvais; ME, Melle; PS, Parcé-sur-Sarthe; RE, Restinclières; SA, Saint-Aulaix-la-Chapelle; SJ, Saint-Jean-d'Angély; SJM, Saint-Jouin-de-Marnes; SM, Saint-Maxire; VER, Verpillères; VZ, Vézénobres

SOC data for the CH, ME, RE, SJ, and VZ sites were taken from Cardinael et al. (2017)

at BE, BO, LB, PS, SJM, and VER sites using a simplified sampling protocol. The SA and SM sites were not sampled for SOC stocks (Table 1). At the BE and PS sites, only the agroforestry plots were sampled for SOC (Table 1), but these sites still remain relevant to study the spatial heterogeneity within agroforestry plots. In total, SOC stocks were therefore measured on 11 agroforestry sites, with nine sites sampled in both agroforestry and control plots.

Soil samples were taken every 10 cm from the surface to a depth of 30 cm using 500-cm^3 cylinders. In the tree rows of silvoarable systems, soil samples were taken at 1 m from a randomly chosen tree and at the half distance between two trees (Fig. 1). In the alleys, soil samples were taken at 1 m from the tree row and in the middle of the alley (Fig. 1). This sampling protocol was repeated three times in the agroforestry plots, around three different trees. In treeless agricultural plots, soil samples were collected at three to six different points spaced of 15 m. At each site, the bulk density (g cm^{-3}) was measured at tree sampling points, one in the tree row, one in the middle of the alley, and one in the control plot.

After air-drying in the lab, soil cores were sieved to 2 mm and weighed without coarse particles > 2 mm. Sub-samples for bulk density determination were oven-dried at 105 °C for 48 h. The bulk density (g cm^{-3}) was calculated as the ratio of the dry mass of fine soil (< 2 mm) to the cylinder volume. Sub-samples for organic C concentrations were dried at 40 °C and ball milled until they passed through a 200-mm mesh sieve. These sub-samples were then analyzed using a CHN elemental analyzer. SOC stocks were calculated on an equivalent soil mass basis (Ellert and Bettany 1995).

Earthworm collection and laboratory analyses

To fulfill the first objective, earthworms were collected at eight silvoarable sites (BE, CH, LB, ME, PS, SA, SJ, VER) in both the agroforestry and the adjacent control plots

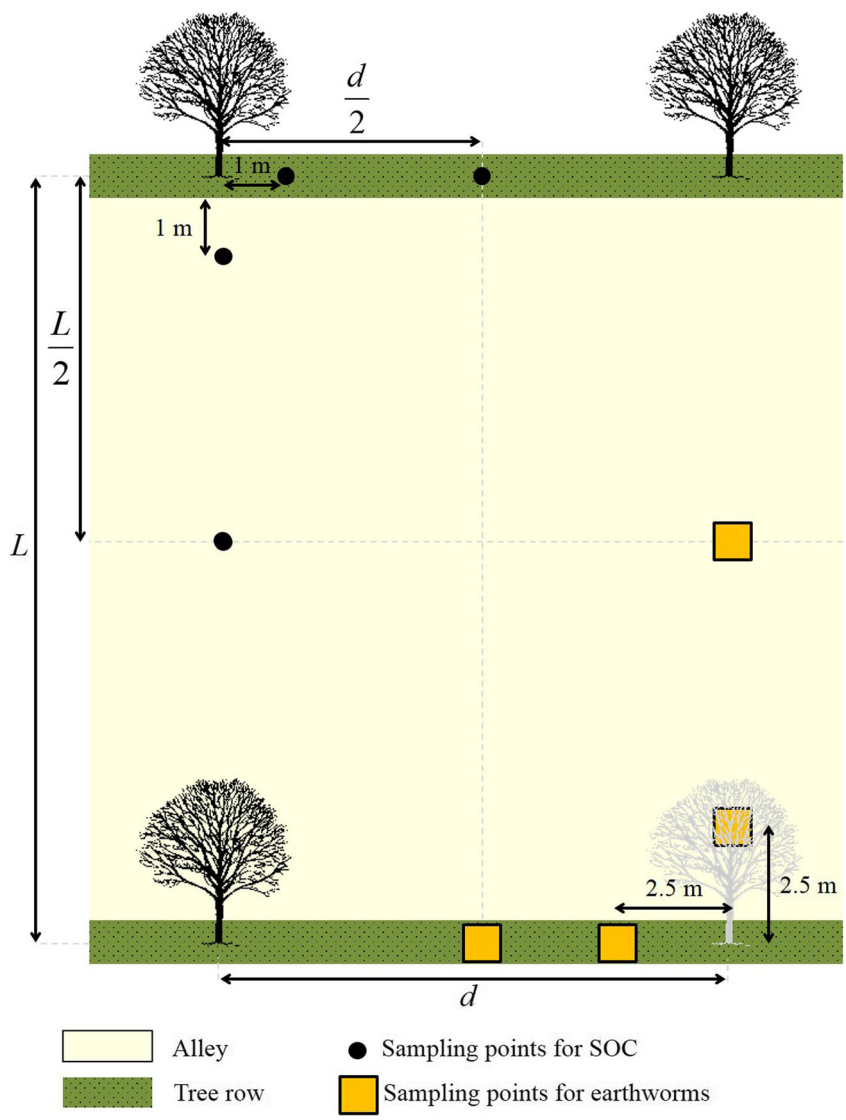

Fig. 1 Sampling design for the silvoarable sites. This sampling protocol for earthworm communities and SOC was repeated three times in the agroforestry plots, around three different trees. L is the distance between tree rows, d is the distance between trees on the rows. SOC: soil organic C

(Table 1). In addition, to complete the second objective, five other sites (BO, RE, SJM, SM, VZ) were also sampled in the agroforestry plot only. Therefore, 13 sites were used to study the spatial heterogeneity of earthworm communities within agroforestry plots.

Earthworms were collected at each site during their maximum biological activity in March/April. In the tree rows of silvoarable systems, earthworms were sampled in the same direction at 2.5 m from a randomly chosen tree and at the mid-distance between two trees (Fig. 1). If spacing between trees along the tree rows was lower than 5 m, only the point at 2.5 m was sampled (it only happened at the LB site). In the alleys, earthworms were sampled at 2.5 m from the tree and in the middle of the alley (Fig. 1). These sampling points were repeated three times (around three different trees). Therefore, earthworms were sampled at six different points per site in the tree rows and alleys. In treeless agricultural plots, earthworms were collected in the middle of the plot, at three different points at 6-m distance.

Earthworms were sampled following the normalized protocol ISO 23611-1, that was modified and validated during the RMQS BioDiv program (Cluzeau et al. 2012). It combined a chemical and manual extraction. Briefly, three watering phases with 10 L of an increasing concentration of formaldehyde (0.25, 0.25, and 0.4%) were applied on 1-m^2 delimited with a frame. After each watering, earthworms were collected during 15 min. To recover earthworms that have not reached the surface, a manual hand sorting inside the sample square was then carried out on a soil cube (25 × 25 × 20 cm, length × width × depth) corresponding to a surface of 1/16 m^2. Earthworms were fixed and stored in formaldehyde (0.4%) at room temperature.

Earthworms were identified to the species level based on morphological criteria (Bouché 1972) and classified into ecological categories defined by Bouché (1972, 1977): epigeic, endogeic, and anecic with a distinction between epi-anecic and strict-anecic earthworms.

Each earthworm individual was then weighted ± 10 mg (fresh formalin weights, full digestive tract) and assigned a stage of development, juvenile, sub-adult, or adult. In total, about 24,000 earthworms were collected, identified at the species level, and weighted. The number of hand sorted earthworms (HS) was multiplied by 16 to estimate the correct number per square meter. It was then added to the number of earthworms counted with the formaldehyde (F) to obtain the total amount of earthworms (FHS, Eq. (1)):

$$\mathrm{FHS} = \mathrm{F} + 16 \times \mathrm{HS} \quad (1)$$

Earthworm diversity was analyzed through three indices, total richness, Shannon index, and the species evenness.

Statistical analysis

One silvoarable site (VZ) associated organic vegetables, while all the other sites concerned arable crops. These sites were analyzed separately. To study the spatial distribution (tree rows vs alleys) of SOC or earthworms within silvoarable sites, we included all sites where the agroforestry plot was measured. When comparing agroforestry and control plots in terms of SOC or earthworms, we only included sites where both plots were sampled. No significant differences were found between SOC stocks and earthworm communities sampled at different distances from the tree, either in the tree rows or in the alleys (data not shown). Samples were therefore combined, and analyses were performed for three distinct modalities: tree rows, alleys, and controls.

We used statistical analyses commonly applied in meta-analysis or experimental network analysis (Makowski et al. 2018). For each location (tree row, alley, control) at each site, we calculated means and standard deviations of the different sampling points for SOC stocks, earthworm total abundance and biomass, earthworm ecological category abundance, total richness, Shannon index, and the species evenness. These means and standard deviations were used to calculate for each site the following effect sizes and their associated confidence intervals. The first one was the ratio of a given variable observed in the tree row to the one observed in the control. The second effect sizes compared alleys to control plots. A last effect size was calculated to study spatial heterogeneity within the agroforestry plot and was the ratio of a given variable observed in the tree row to the one observed in the alley. A logarithm conversion was then applied to each effect size to center the values around 0. In order to estimate the mean effect size and its confidence interval across sites, we applied a random-effect model and a restricted maximum likelihood (REML) approach the using the *nlme* package (Pinheiro et al. 2013). Each site was considered as a random factor, making the results obtained on this experimental network more generalizable.

For the earthworm individual biomass, we only selected sites where species were represented by at least three individual adults in at least two of the three locations (tree row, alley, and agricultural plot). Differences in earthworm individual biomass were compared species by species, location by location, and site by site. As the individual biomass of earthworms depends on each species, it is not relevant to compare absolute values between locations and sites. The individual weight of each earthworm species in the alleys and control at a given site was compared to the weight of same species in the tree rows and expressed as a relative weight. The difference relative weights of each species were then averaged for each ecological category, location, and site. For each earthworm ecological category, we used a linear mixed-effects model using *lme4* package (Bates 2010) with sites as a random factor, followed

by Tukey HSD tests for post hoc pairwise comparisons to test differences in earthworm individual biomass between tree rows, alleys, and control plots. The full list of earthworm species identified per site and location is available in Table S1 in the Supplementary materials.

Statistical analyses were performed using R software, version 3.1.1 (R Development Core Team 2013). Significance was evaluated in all cases at p value < 0.05.

Results

Agroforestry effect on SOC stocks

The SOC stocks varied greatly between sites, reflecting various soil types and historical management (Fig. 2). Mean SOC stocks in 0–30 cm were significantly higher in tree rows compared to those in control plots across the sites (p value = 0.004). No significant difference was found between alleys and control plots across the sites (p value = 0.07). Mean SOC stocks in 0–30 cm were significantly higher in the tree rows than those in the alleys across the silvoarable sites (p value = 0.003) (Fig. 3). Site-specific log effect sizes for SOC can be seen in the Supplementary materials (Fig. S1).

Across the nine silvoarable sites (average age 12.1 years) where the three modalities where sampled (BO, CH, LB, ME, PS, RE, SJ, SJM, VER) for SOC, the mean (± 95% confidence intervals) delta SOC stocks were 19.46 ± 13.77 and 10.66 ± 10.35 t C ha^{-1} between tree rows and control plot and between alleys and control plot, respectively.

Total earthworm abundance and biomass

Total earthworm abundance varied a lot between sites (Fig. 4). In general, the total earthworm abundance followed the same patterns across the silvoarable sites: tree rows ≫ alleys ≥ controls. At the VZ site (organic vegetables), earthworm total abundance was higher in the alleys than those in the tree rows.

For the eight silvoarable sites where the three modalities (control, alley, tree row) were sampled, the mean total earthworm abundances and associated 95% confidence intervals were 238 ± 124, 289 ± 85, and 595 ± 168 individuals m^{-2} in the control, alleys, and tree rows of silvoarable sites, respectively (Fig. 5). Mean total earthworm abundance in the tree rows was therefore 150% higher than that in the control, 106% than that in the alleys, and earthworm abundance in the alleys was 21% higher than that in the control. The corresponding mean total earthworm biomasses and associated 95% confidence intervals were 77 ± 44, 96 ± 34, and 152 ± 58 g m^{-2}, respectively (Fig. 5).

Mean earthworm total abundance and total biomass were not significantly different between alleys and control plots across the sites (p value = 0.55 and 0.29, respectively). However, they were significantly higher in the tree rows than those in the control plots across all sites (p value = 0.007 and 0.008, respectively) (Fig. 3). Mean earthworm total abundance and total biomass were also significantly higher in the tree rows than those in the alleys across the agroforestry sites (p value < 0.001 and p value = 0.003, respectively) (Fig. 3). Site-specific effects are detailed in the Supplementary materials (Fig. S2).

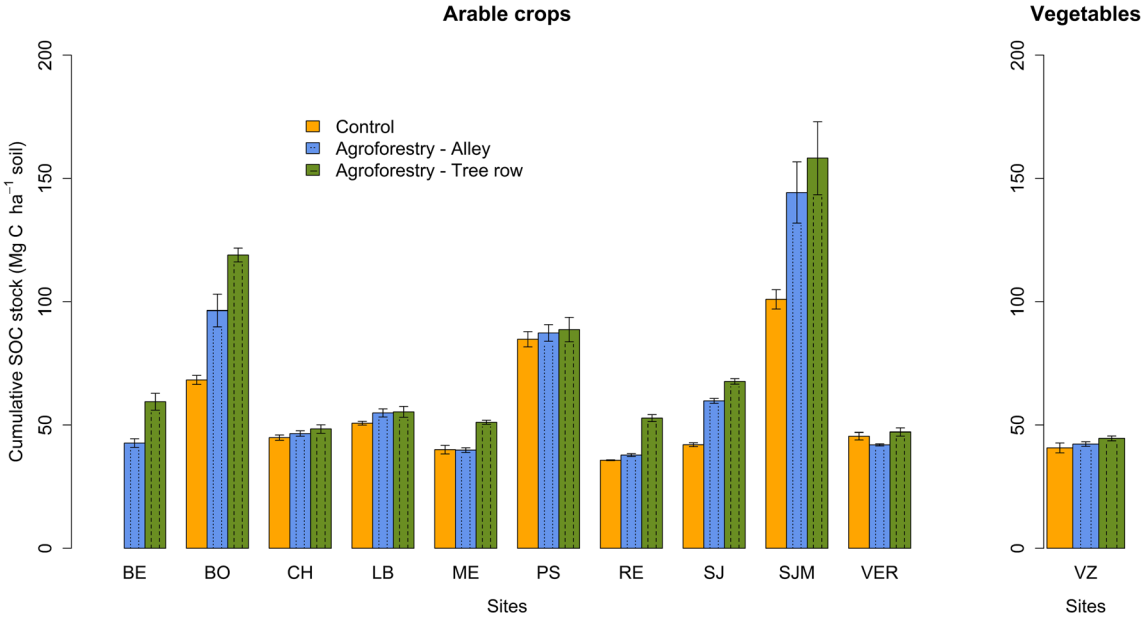

Fig. 2 Soil organic C stock (Mg C ha^{-1}) in 0–30 cm at the different agroforestry sites. Error bars represent standard errors

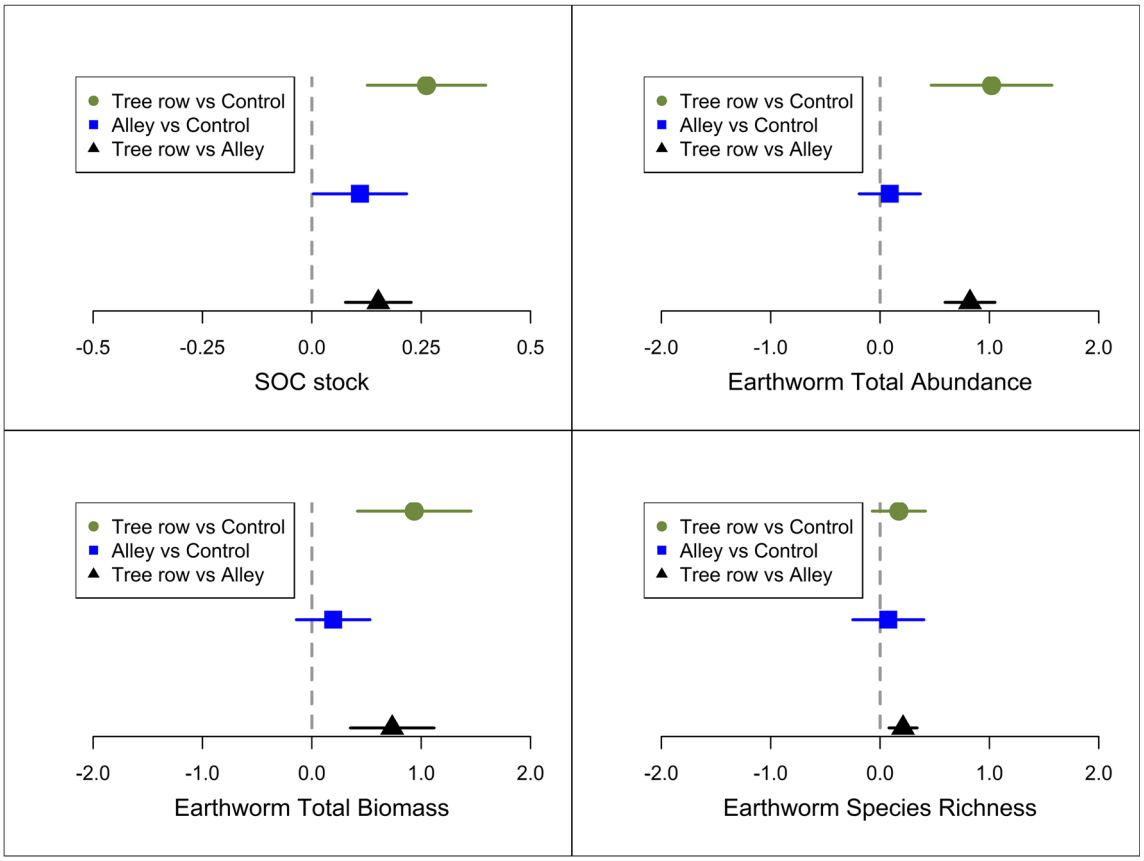

Fig. 3 Mean log effect sizes of SOC stocks (0–30 cm), earthworm total abundance and biomass, and earthworm species richness between tree rows and control plots, alleys and control plots, and tree rows and alleys at the different silvoarable sites (excluding the VZ site with organic vegetables). Error bars represent 95% confidence intervals

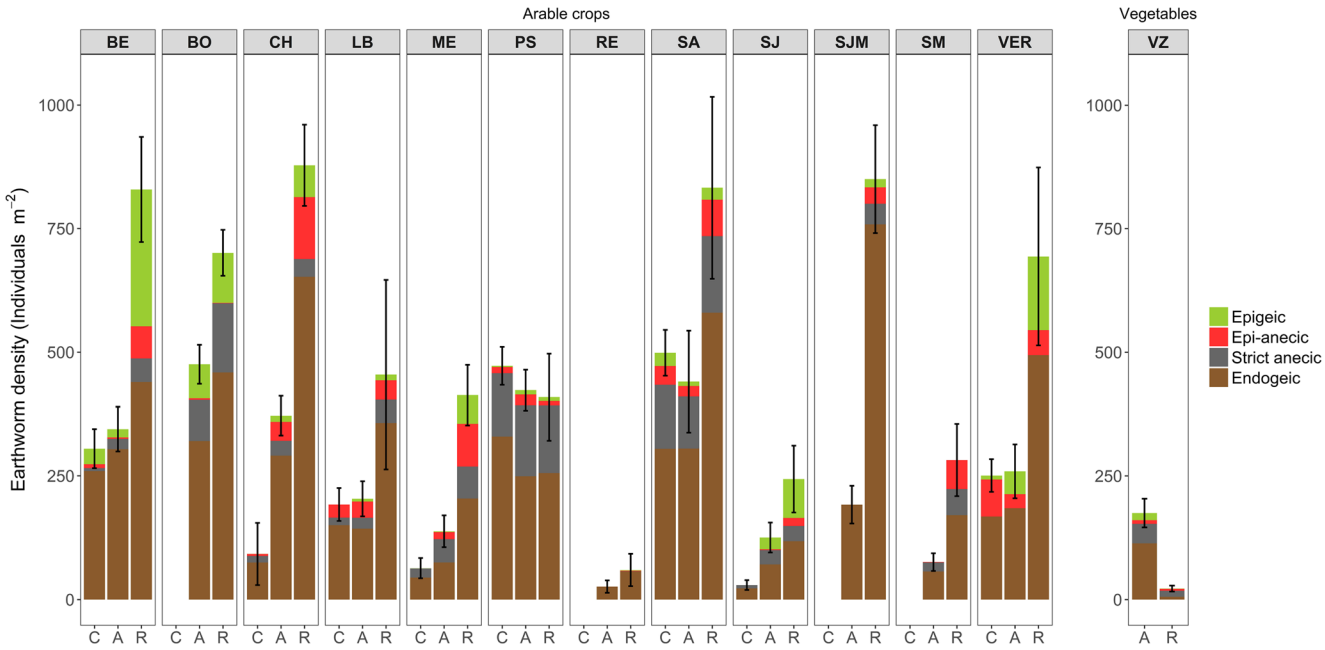

Fig. 4 Distribution of earthworm mean total abundance in the four ecological categories at the different agroforestry sites and modalities. C, control; A, alley; R, tree row. Error bars represent standard errors for total earthworm abundance. For the sites BO, RE, SJM, SM, and VZ, the control plots were not sampled for earthworms (see Table 1)

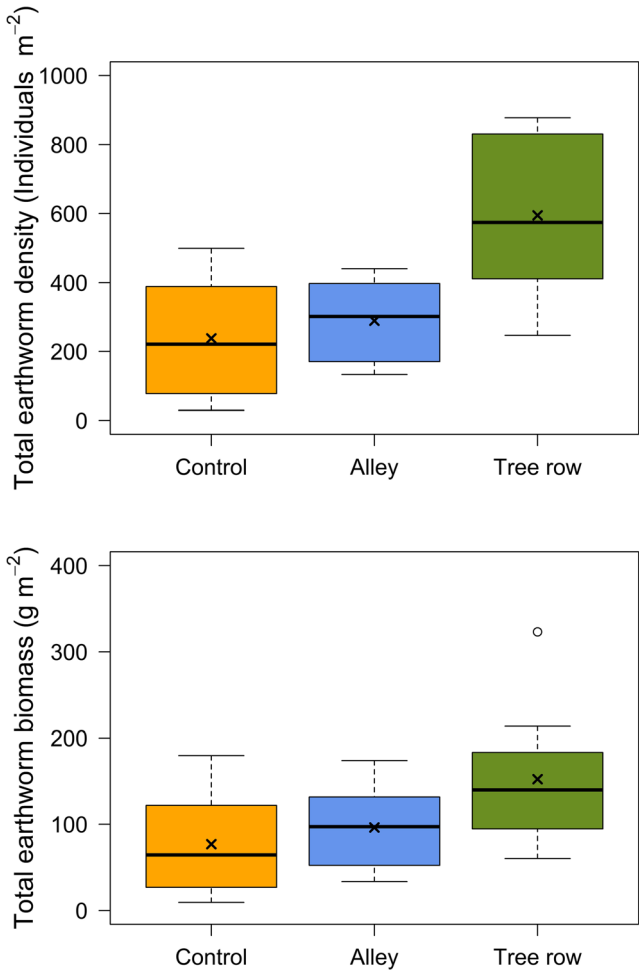

Fig. 5 Total earthworm abundance and biomass in the control, alleys, and tree rows of the eight silvoarable sites where the three modalities (control, alley, tree rows) were sampled. The following sites are concerned: BE, CH, LB, ME, PS, SA, SJ, and VER. Upper and lower edges of boxes indicate 75th and 25th percentiles, horizontal lines within boxes indicate median, whiskers below and above the boxes indicate the 10th and 90th percentiles, and crosses indicate arithmetic means. Outliers are plotted as individual points

Earthworm ecological categories

A total of 20 species of Lumbricidae were identified across the agroforestry sites, with several representatives of the four ecological categories (Table S1). Total abundance was largely dominated by endogeic species such as *Allolobophora chlorotica chlorotica* or *Aporrectodea caliginosa caliginosa*, whereas total biomass was mainly driven by large-sized anecics (such as *Aporrectodea longa*).

Endogeic were the most abundant earthworms across all sites (Fig. 4). Across the eight silvoarable sites (BE, CH, LB, ME, PS, SA, SJ, VER) where the three modalities were sampled, mean endogeic abundances and associated 95% confidence intervals were 169.2 ± 82.3, 202.7 ± 68.6, and 387.6 ± 130.0 ind m^{-2} in the controls, alleys, and tree rows, respectively. Mean epigeic abundances and associated 95% confidence intervals were 8.6 ± 9.0, 15.5 ± 9.9, and 83.8 ± 62.6 ind m^{-2} in the controls, alleys, and tree rows, respectively. The abundance of epigeic, strict anecic, and endogeic was significantly higher in the tree rows than that in the control plots across the silvoarable sites (p values = 0.02, 0.006, and 0.009, respectively) (Fig. S3), but no difference was observed for the epi-anecic (p value = 0.16). In the alleys, only the abundance of the strict anecic was significantly higher than those in the control plots (p value = 0.03). Within silvoarable plots, the mean abundance of each earthworm ecological category was significantly higher in the tree rows than those in the alleys (p values = 0.01, 0.009, 0.007, and < 0.0001 for epigeic, strict anecic, epi-anecic, and endogeic earthworms, respectively) (Fig. S3).

Earthworm diversity indicators

Across the eight silvoarable sites (BE, CH, LB, ME, PS, SA, SJ, VER) where the three modalities were sampled, earthworm species richness was 5.3, 6.1, and 7.3 for the controls, alleys, and tree rows, respectively. This indicator was significantly higher in the tree rows than that in the alleys within agroforestry plots (Fig. 3). However, it was not significantly different either between tree rows and control plots or between alleys and control plots across the sites (Fig. 3). The same results were obtained for the Shannon index and the species evenness (Fig. S4).

Earthworm individual weight

The ANOVA revealed that the earthworm individual weight did not significantly vary across sites (p value = 0.629). However, the individual weight significantly depended on the location (p value < 0.001) and on the ecological category (p value = 0.002). The interaction between the location and the ecological category was also significant (p value < 0.001). Adult individual mean weights of epigeic, epic-anecic, strict anecic, and epigeic species ranged from 0.05 to 0.43 g, 0.41 to 8.80 g, 0.84 to 5.78 g, and 0.03 to 0.89 g, respectively. An example of earthworm individual weight across the sites and locations is shown in Fig. S5 for *A. chlorotica chlorotica* species.

Adult individual mean weights of epigeic and epi-anecic species were significantly higher in the controls (+ 26 and + 27%, respectively) and in the alleys (+ 14 and + 13%, respectively) compared to those in the tree rows (Fig. 6). Adult individual mean weight of endogeic adults was significantly higher in the alleys and in the controls (+ 6 and + 7%, respectively) than those in the tree rows (Fig. 6). Surprisingly, there was no difference between the adult individual weight of strict anecic species in the tree rows and the alleys (+ 6%), but they were significantly lighter than that in the controls.

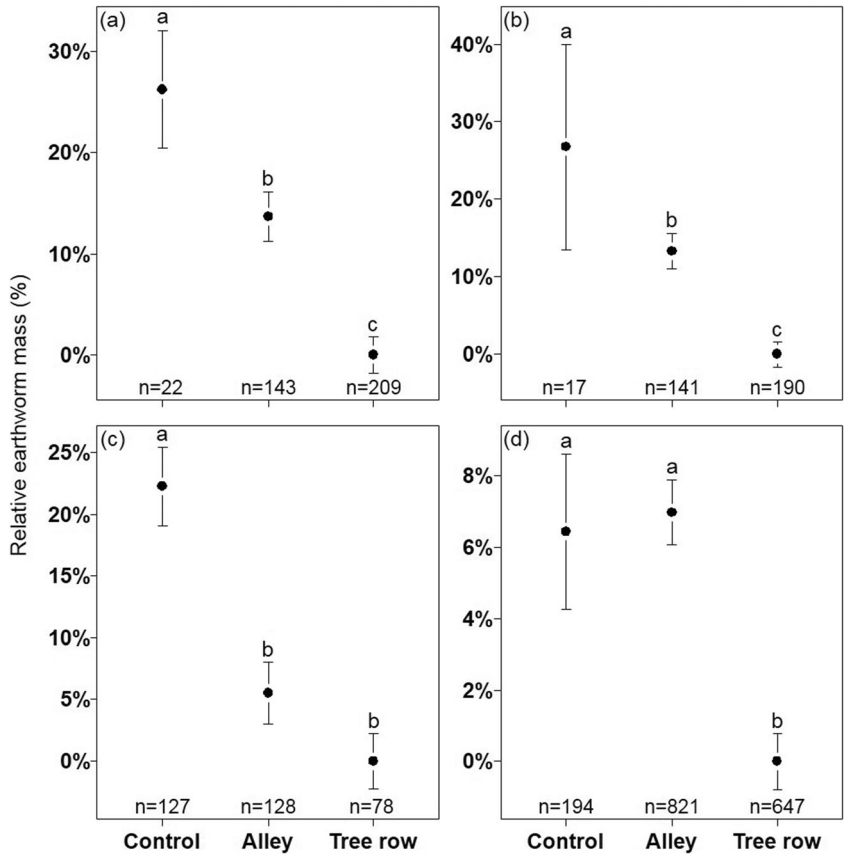

Fig. 6 Relative adult earthworm weights in controls and alleys compared to those in tree rows according to the ecological category: **a** epigeic, **b** epi-anecic, **c** strict anecic, and **d** endogeic. Values are arithmetic means, and error bars represent standard errors; n: number of individuals. Different letters denote significant differences among relative adult earthworm weight with a > b > c (Tukey test results)

Discussion

Spatial variation of SOC stocks

We found that SOC stocks were not significantly higher (p value = 0.07) in the alleys than that in control plots across the sites. This is probably the result of the high percentage of young sites within this experimental network (three sites older than 15 years, ten sites younger than 10 years). However, this study confirms previous results showing that SOC stocks in silvoarable systems are significantly higher in the tree rows than that in the alleys and control plots (Bambrick et al. 2010; Cardinael et al. 2015a, 2017). Cardinael et al. (2018b) found that in an 18-year-old silvoarable system (110 trees ha^{-1}), tree rows received two times more organic C inputs compared to the control plot and 65% more than alleys. The additional organic inputs came from litterfall, from tree root mortality (Cardinael et al. 2015b; Germon et al. 2016), and also from the herbaceous vegetation growing between the trees (Cardinael et al. 2018b). In this study, SOC stocks in young silvoarable plots were found to be increased mainly in the tree rows, which could be ascribed to this herbaceous vegetation, analogous to grassland strips. Finally, tree rows are usually untilled, and a higher aggregate stability could contribute to SOC sequestration (Udawatta et al. 2008).

Higher earthworm abundance, biomass, and richness in tree rows

In general, total earthworm abundance, biomass, and richness were higher in the tree rows than those in the alleys and in the control plots. Only the abundance of epi-anecic earthworms and the richness between tree rows and control plots were not different. These results are similar to those observed in grassy or hedge field-margins of arable fields. As previously observed by Smith et al. (2008a), Nieminen et al. (2011), Roarty and Schmidt (2013) and Crittenden et al. (2015), earthworm abundance, biomass, and richness were often higher in field margin than those in the agricultural field. It is well-known that soil disturbance, food supply, and soil physical properties like soil moisture and compaction affect earthworm communities (Lee 1985; Curry 1998; Chan 2001; Capowiez et al. 2009; Pelosi et al. 2014; van Capelle et al. 2012). It is therefore likely that the greater abundance, biomass, and diversity in the tree rows were the result of the absence of cultivation and the provision of food resources as observed in grasslands compared to croplands (Curry 1998). However, with some agricultural practices such as manure addition, diversified rotations, and reduced tillage, Lagerlöf et al. (2002) observed higher abundance in cropland than that in field margins. Hof and Bright (2010) observed opposite results with

a higher earthworm total abundance and biomass in croplands at 20 m from a grassy field margin than those from without grassy field margin. At the VZ silvoarable site associating hybrid walnut trees with organic vegetables, alleys were usually amended with poplar ramial chipped wood. This increase in the trophic resource for earthworms could explain their higher abundance in the alley than those in the tree row at this site. Pérès et al. (1998) also found an increase in earthworm abundance after application of fresh poplar bark. In addition, the PS site was managed under no-tillage practices, which could explain the similar earthworm abundance between the alleys and tree rows (Chan 2001). Inter-site variability could also explain others factors like soil characteristics, tree density and age, tree species composition, tree row plant community characteristics, and management of the tree row (sown or spontaneous vegetation, mowing …), but due to a lack of data, we were unable to test it. Tree species composition and herbaceous tree row plant community could indeed result in quantitatively and qualitatively different litter inputs and varying plant effects on belowground conditions because of different root densities and characteristics.

Are tree rows an earthworm source for the cropped alleys?

In our study, earthworm communities were not very different between the alleys and the control plots. Moreover, no difference was found between the earthworm abundance collected in the alleys close to the tree rows or in the middle of the alleys. This finding is similar to results observed with field margins, suggesting that tree rows are a favorable habitat but do not serve as a source of earthworms for the alleys. Despite a higher earthworm abundance, biomass, or richness in the field margins, Roarty and Schmidt (2013) and Crittenden et al. (2015) did not observe an increase of in-field earthworm populations as usually documented for more mobile, aboveground invertebrate taxa through colonization or spill-over effects. They therefore suggested that earthworm populations inside agricultural fields rely on residual, surviving in-field populations, rather than on immigration from surrounding land. Nevertheless, earthworms are also mobile species, their dispersal rate and distance vary between groups of earthworms (Eijsackers 2011), but this mobility is affected by other factors, such as agricultural practices (Nieminen et al. 2011) or climatic conditions (Cluzeau 1992). Nuutinen et al. (2011) found that margins could have importance as a source for epi-anecics when they invade a new arable habitat. The complexity of earthworm dispersion dynamics between the tree row and the alley could be further studied using molecular tools to distinguish subpopulations (Mathieu et al. 2010; Dupont et al. 2015, 2017) or earthworm tagging (Butt et al. 2009; Mathieu et al. 2018). Some studies have shown a earthworm species-related mobility, from 4.5 m per year for *Lumbricus terrestris* (Hoogerkamp et al. 1983), 5.9–6.7 m per year for *A. longa* (Schon et al. 2014), and up to 6.3 m per year for all the earthworm community (Ligthart and Peek 1997). Zeithaml et al. (2009) and Hof and Bright (2010) observed an abundance of earthworms greater at 20 or 25 m from the field margin than those at 5 or 10 m from the field margin, but the reasons are not clear. Our results also revealed that strict-anecic earthworm abundance was overall higher in alleys than those in control plots. This ecological category could potentially benefit from the local microclimate and organic matter brought by trees in the agroforestry system. Strict-anecic earthworms are geo-saprophagous, meaning that they consume not only highly humified soil organic but also decaying organic matter and could benefit from increased SOC stocks (Bouché and Kretzschmar 1974; Kretzschmar 1977; Ferrière 1980). In a previous study, Cardinael et al. (2015a) found that most of the additional SOC in agroforestry compared to those in control plots was made by decaying particulate organic matter (200–2000 and 50–200 μm). This type of organic matter might be more palatable for strict-anecic than for endogeic earthworms, which are more geophagous, and explain why the abundance of endogeic earthworm was not enhanced in alleys compared to those in control plots.

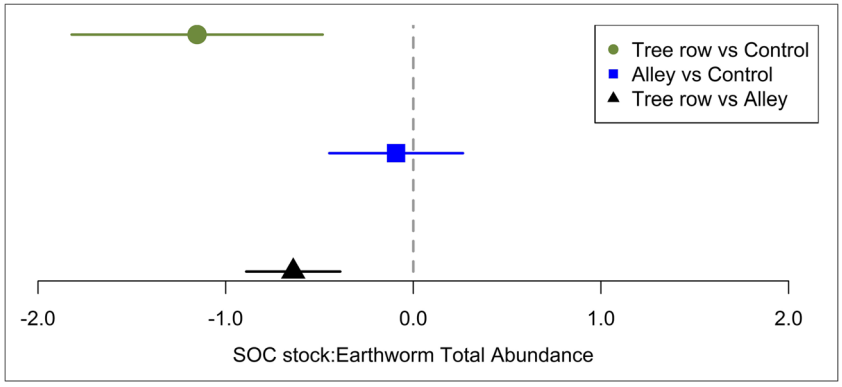

Fig. 7 Log effect size of the ratio between SOC stocks (0–30 cm) and total earthworm biomass. Error bars represent 95% confidence intervals

Effect of earthworm density on individual weight

Within each ecological category, earthworm individual biomass was at the lowest in the tree rows. Tree rows were characterized by a high density of earthworms compared to those in the control plots. A similar pattern could be observed to a lesser extent between the tree row and the alley, as epigeic, epi-anecic, and endogeic individual biomass were also smaller in the tree rows compared to those in the alley. The negative relationship between earthworm abundance and individual biomass, observed in our study, has already been shown in laboratory experiment with different earthworm density gradients (Neuhauser et al. 1980; Hartenstein and Amico 1983; Butt et al. 1994), but to our knowledge, this is the first study revealing this phenomenon in situ. Butt et al. (1994) suggested that this density effect was not a function of food deprivation as excess feed was provided. Here, despite increased SOC stocks, the amount of available total organic C per earthworm individual was lower in the tree rows than those in the control plots due to their higher total abundance (Figs. 7, S6, and S7). This suggests a higher intra- and/or inter-specific competition in the tree rows. Similarly to what we observed in the tree rows, Butt et al. (1994) also showed in a laboratory experiment that a larger density of *L. terrestris* had a negative effect on growth rate and on final earthworm individual weight, but resulted in a greater total biomass. A lower earthworm individual weight could potentially impact not only their provision of ecosystem services but also the population dynamics (Lavelle 1983). For instance, Hoeffner et al. (2018) observed that surface litter incorporation by epi-anecic earthworms was not dependent on the earthworm species but was highly correlated to the initial mean weight of earthworms.

Conclusions

This study showed that earthworm communities and SOC stocks were modified in agroforestry systems. More precisely, we found that earthworm total biomass and abundance were much higher in the tree rows than that in the cropped alleys or than that in the control plots. A similar result was observed for SOC stocks. However, tree rows did not seem to serve as a source of earthworms for the alleys. Finally, despite higher SOC stocks, adult earthworm individual weights were found to be lower in the tree rows than those in the cropped alleys, probably due to intra- and/or inter-specific competition. Further studies are needed to assess how earthworm activities and subsequent ecosystem services could be modified in agroforestry systems.

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References

Araujo Y, López-Hernández D (1999) Earthworm populations in a savanna-agroforestry system of Venezuelan Amazonia. Biol Fertil Soils 29:413–418

Bambrick AD, Whalen JK, Bradley RL, Cogliastro A, Gordon AM, Olivier A, Thevathasan NV (2010) Spatial heterogeneity of soil organic carbon in tree-based intercropping systems in Quebec and Ontario, Canada. Agrofor Syst 79:343–353

Bates DM (2010) lme4: mixed-effects modeling with R. Springer, 145p. URL lme4.r-forge.rproject.org/book/front.pdf

Bertrand M, Barot S, Blouin M, Whalen J, de Oliveira T, Roger-Estrade J (2015) Earthworm services for cropping systems. A review. Agron Sustain Dev 35:553–567

Blouin M, Hodson ME, Delgado EA, Baker G, Brussaard L, Butt KR, Dai J, Denddoven L, Pérès G, Tondoh JE, Cluzeau D, Brun JJ (2013) A review of earthworm impact on soil function and ecosystem services. Eur J Soil Sci 64:161–182

Bouché MB (1972) Lombriciens de France: écologie et systématique. Institut National de la Recherche Agronomique, Paris

Bouché MB (1977) Stratégies lombriciennes. Ecol Bull 25:122–132

Bouché MB, Kretzschmar A (1974) Fonctions des lombriciens II. Recherches méthodologiques pour l'analyse qualitative de la matière organique végétale ingérée (étude du peuplement de la station R.C.P.-165/P.B.I). Rev Ecol Biol Sol 11:127–139

Briones MJI, Schmidt O (2017) Conventional tillage decreases the abundance and biomass of earthworms and alters their community structure in a global meta-analysis. Glob Chang Biol 23:4396–4419

Butt KR, Frederickson J, Morris RM (1994) Effect of earthworm density on the growth and reproduction of *Lumbricus terrestris* L. (Oligochaeta: Lumbricidae) in culture. Pedobiologia 38:254–261

Butt KR, Briones MJI, Lowe CN (2009) Is tagging with visual implant elastomer a reliable technique for marking earthworms? Pesq Agropec Bras 44:969–974

Capowiez Y, Cadoux S, Bouchant P, Ruy S, Roger-Estrade J, Richard G, Boizard H (2009) The effect of tillage type and cropping system on earthworm communities, macroporosity and water infiltration. Soil Tillage Res 105:209–216

Cardinael R, Chevallier T, Barthès BG, Saby NPA, Parent T, Dupraz C, Bernoux M, Chenu C (2015a) Impact of alley cropping agroforestry on stocks, forms and spatial distribution of soil organic carbon—a case study in a Mediterranean context. Geoderma 259–260:288–299

Cardinael R, Mao Z, Prieto I, Stokes A, Dupraz C, Kim JH, Jourdan C (2015b) Competition with winter crops induces deeper rooting of walnut trees in a Mediterranean alley cropping agroforestry system. Plant Soil 391:219–235

Cardinael R, Chevallier T, Cambou A, Béral C, Barthès BG, Dupraz C, Durand C, Kouakoua E, Chenu C (2017) Increased soil organic

carbon stocks under agroforestry: a survey of six different sites in France. Agric Ecosyst Environ 236:243–255

Cardinael R, Umulisa V, Toudert A, Olivier A, Bockel L, Bernoux M (2018a) Revisiting IPCC Tier 1 coefficients for soil organic and biomass carbon storage in agroforestry systems. Environ Res Lett. https://doi.org/10.1088/1748-9326/aaeb5f in press

Cardinael R, Guenet B, Chevallier T, Dupraz C, Cozzi T, Chenu C (2018b) High organic inputs explain shallow and deep SOC storage in a long-term agroforestry system—combining experimental and modeling approaches. Biogeosciences 15:297–317

Chan KY (2001) An overview of some tillage impacts on earthworm population abundance and diversity—implications for functioning in soils. Soil Tillage Res 57:179–191

Chatterjee N, Nair PKR, Chakraborty S, Nair VD (2018) Changes in soil carbon stocks across the Forest-Agroforest-Agriculture/Pasture continuum in various agroecological regions: a meta-analysis. Agric Ecosyst Environ 266:55–67

Cluzeau D (1992) Structure et dynamique des peuplements lombriciens dans des systèmes tempérés anthropisés. Université Rennes 1, Rennes

Cluzeau D, Guernion M, Chaussod R, Martin-Laurent F, Villenave C, Cortet J, Ruiz-Camacho N, Pernin C, Mateille T, Philippot L, Bellido A, Rougé L, Arrouays D, Bispo A, Pérès G (2012) Integration of biodiversity in soil quality monitoring: baselines for microbial and soil fauna parameters for different land-use types. Eur J Soil Biol 49:63–72

Crittenden SJ, Huerta E, de Goede RGM, Pulleman MM (2015) Earthworm assemblages as affected by field margin strips and tillage intensity: an on-farm approach. Eur J Soil Biol 66:49–56

Curry J (1998) Factors affecting earthworm abundance in soils. In: Edwards CA (ed) Earthworm ecology. St. Lucie Press, Raton, pp 37–64

de Stefano A, Jacobson MG (2018) Soil carbon sequestration in agroforestry systems: a meta-analysis. Agrofor Syst 92:285–299

Decaëns T, Bureau F, Margerie P (2003) Earthworm communities in a wet agricultural landscape of the Seine Valley (Upper Normandy, France): the 7th International Symposium on Earthworm Ecology Cardiff Wales 2002. Pedobiologia 47:479–489

Decaëns T, Margerie P, Aubert M, Hedde M, Bureau F (2008) Assembly rules within earthworm communities in North-Western France—a regional analysis. Appl Soil Ecol 39:321–335

Dupont L, Gresille Y, Richard B, Decaëns T, Mathieu J (2015) Dispersal constraints and fine-scale spatial genetic structure in two earthworm species. Biol J Linn Soc 114:335–347

Dupont L, Torres-Leguizamon M, René-Corail P, Mathieu J (2017) Landscape features impact connectivity between soil populations: a comparative study of gene flow in earthworms. Mol Ecol 26: 3128–3140

Eijsackers H (2011) Earthworms as colonizers of natural and cultivated soil environments. Appl Soil Ecol 50:1–13

Ellert BH, Bettany JR (1995) Calculation of organic matter and nutrients stored in soils under contrasting management regimes. Can J Soil Sci 75:529–538

Feliciano D, Ledo A, Hillier J, Nayak DR (2018) Which agroforestry options give the greatest soil and above ground carbon benefits in different world regions? Agric Ecosyst Environ 254:117–129

Ferrière G (1980) Fonctions des Lombriciens. VII. Une méthode d'analyse de la matière organique végétale ingérée. Pedobiologia 20:263–273

Fonte SJ, Barrios E, Six J (2010) Earthworms, soil fertility and aggregate-associated soil organic matter dynamics in the Quesungual agroforestry system. Geoderma 155:320–328

Frazão J, de Goede RGM, Brussaard L, Faber JH, Groot JCJ, Pulleman MM (2017) Earthworm communities in arable fields and restored field margins, as related to management practices and surrounding landscape diversity. Agric Ecosyst Environ 248:1–8

Frouz J, Pižl V, Cienciala E, Kalčík J (2009) Carbon storage in post-mining forest soil, the role of tree biomass and soil bioturbation. Biogeochemistry 94:111–121

Germon A, Cardinael R, Prieto I, Mao Z, Kim JH, Stokes A, Dupraz C, Laclau JP, Jourdan C (2016) Unexpected phenology and lifespan of shallow and deep fine roots of walnut trees grown in a silvoarable Mediterranean agroforestry system. Plant Soil 401:409–426

Hartenstein R, Amico L (1983) Production and carrying for the earthworm *Lumbricus terrestris* in culture. Soil Biol Biochem 15:51–54

Hauser S (1993) Distribution and activity of earthworms and contribution to nutrient recycling in alley cropping. Biol Fertil Soils 15:16–20

Hauser S, Osawalam DO, Vanlauwe B (1998) Spatial and temporal gradients of earthworms casting activity in alley cropping systems. Agrofor Syst 41:127–137

Hmar L, Ramanujam SN (2014) Earthworm cast production and physico-chemical properties in two agroforestry systems of Mizoram (India). Trop Ecol 55:75–84

Hoeffner K, Monard C, Santonja M, Cluzeau D (2018) Feeding behaviour of epi-anecic earthworm species and their impacts on soil microbial communities. Soil Biol Biochem 125:1–9

Hof AR, Bright PW (2010) The impact of grassy field margins on macroinvertebrate abundance in adjacent arable fields. Agric Ecosyst Environ 139:280–283

Hoogerkamp M, Rogaar H, Eijsackers HJP (1983) Effect of earthworms on grassland on recently reclaimed polder soils in the Netherlands. In: Satchell JE (ed) Earthworm ecology. Springer, Dordrecht, pp 85–105

Jégou D, Cluzeau D, Balesdent J, Tréhen P (1998) Effects of four ecological categories of earthworms on carbon transfer in soil. Appl Soil Ecol 9:249–255

Jégou D, Capowiez Y, Cluzeau D (2001a) Interactions between earthworm species in artificial soil cores assessed through the 3D reconstruction of the burrow systems. Geoderma 102:123–137

Jégou D, Schrader S, Diestel H, Cluzeau D (2001b) Morphological, physical and biochemical characteristics of burrow walls formed by earthworms. Appl Soil Ecol 17:165–174

Jouquet P, Dauber J, Lagerlöf J, Lavelle P, Lepage M (2006) Soil invertebrates as ecosystem engineers: intended and accidental effects on soil and feedback loops. Appl Soil Ecol 32:153–164

Kretzschmar A (1977) Etude du transit intestinal des Lombriciens aneciques. II. Resultats et interpretation ecologique. Ecol Bull 25: 210–221

Lagerlöf J, Goffre B, Vincent C (2002) The importance of field boundaries for earthworms (Lumbricidae) in the Swedish agricultural landscape. Agric Ecosyst Environ 89:91–103

Lavelle P (1983) The structure of earthworm communities. In: Satchell J (ed) Earthworm ecology. Chapman & Hall, London, p 449–466

Lavelle P (1988) Earthworm activities and the soil system. Biol Fertil Soils 6:237–251

Lavelle P (1997) Faunal activities and soil processes: adaptative strategy that determine ecosystem function. Adv Ecol Res 27:93–132

Lavelle P, Decaëns T, Aubert M, Barot S, Blouin M, Bureau F, Margerie P, Mora P, Rossi JP (2006) Soil invertebrates and ecosystem services. Eur J Soil Biol 42:3–15

Lee KE (1985) Earthworms their ecology and relationships with soils and land use. Academic Press, Sydney

Ligthart TN, Peek GJCW (1997) Evolution of earthworm burrow systems after inoculation of lumbricid earthworms in a pasture in the Netherlands. Soil Biol Biochem 29:453–462

Makowski D, Piraux F, Brun F (2018) De l'analyse des réseaux expérimentaux à la méta-analyse. Méthodes et applications avec le logiciel R pour les sciences agronomiques et environnementales. EditionsQuae, p 162

Mathieu J, Barot S, Blouin M, Caro G, Decaëns T, Dubs F, Dupont L, Jouquet P, Nai P (2010) Habitat quality, conspecific density, and habitat pre-use affect the dispersal behaviour of two earthworm

species, *Aporrectodea icterica* and *Dendrobaena veneta*, in a mesocosm experiment. Soil Biol Biochem 42:203–209

Mathieu J, Caro G, Dupont L (2018) Methods for studying earthworm dispersal. Appl Soil Ecol 123:339–344

Nair PKR (1993) An introduction to agroforestry. Kluwer Academic Publishers, Dordrecht

Neuhauser E, Hartenstein R, Kaplan D (1980) Growth of the earthworm *Eisenia foetida* in relation to population density and food rationing. Oikos 35:93–98

Nieminen M, Ketoja E, Mikola J, Terhivuo J, Siren T, Nuutinen V (2011) Local land use effects and regional environmental limits on earthworm communities in Finnish arable landscapes. Ecol Appl 21: 3162–3177

Nuutinen V, Butt KR, Jauhiainen L (2011) Field margins and management affect settlement and spread of an introduced dew-worm (*Lumbricus terrestris* L.) population. Pedobiologia 54S:S167–S172

Pelosi C, Bertrand M, Roger-Estrade J (2009) Earthworm community in conventional, organic and direct seeding with living mulch cropping systems. Agron Sustain Dev 29:287–295

Pelosi C, Pey B, Hedde M, Caro G, Capowiez Y, Guernion M, Peigné J, Diron D, Bertrand M, Cluzeau D (2014) Reducing tillage in cultivated fields increases earthworm functional diversity. Appl Soil Ecol 83:79–87

Pérès G, Cluzeau D, Curmi P, Hallaire V (1998) Earthworm activity and soil structure changes due to organic enrichments in vineyard systems. Biol Fertil Soils 27:417–424

Pinheiro J, Bates D, DebRoy S, Sarkar D, R Development Core Team (2013) nlme: linear and nonlinear mixed effects models. R package version 3.1–111

Ponge JF, Pérès G, Guernion M, Ruiz-Camacho N, Cortet J, Pernin C, Villenave C, Chaussod R, Martin-Laurent F, Bispo A, Cluzeau D (2013) The impact of agricultural practices on soil biota: a regional study. Soil Biol Biochem 67:271–284

Potvin LR, Lilleskov EA (2017) Introduced earthworm species exhibited unique patterns of seasonal activity and vertical distribution, and *Lumbricus terrestris* burrows remained usable for at least 7 years in hardwood and pine stands. Biol Fertil Soils 53:187–198

Price GW, Gordon AM (1999) Spatial and temporal distribution of earthworms in a temperate intercropping system in southern Ontario, Canada. Agrofor Syst 44:141–149

R Development Core Team (2013) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL http://www.R-project.org/

Roarty S, Schmidt O (2013) Permanent and new arable field margins support large earthworm communities but do not increase in-field populations. Agric Ecosyst Environ 170:45–55

Saharan N, Singh RP (1988) Influence of earthworm activity on soil fertility in agroforestry systems. Ann Arid Zone 27:81–85

Schon NL, Mackay AD, Gray RA, Dodd MB (2014) The action of an anecic earthworm (*Aporrectodea longa*) on vertical soil carbon distribution in New Zealand pastures several decades after their introduction. Eur J Soil Biol 62:101–104

Shi L, Feng W, Xu J, Kuzyakov Y (2018) Agroforestry systems: meta-analysis of soil carbon stocks, sequestration processes, and future potentials. Land Degrad Dev:1–12

Shipitalo MJ, Protz R, Tomlin AD (1988) Effect of diet on the feeding and casting activity of *Lumbricus terrestris* and *Lumbricus rubellus* in laboratory culture. Soil Biol Biochem 20:233–237

Smith J, Potts SG, Woodcock BA, Eggleton P (2008a) Can arable field margins be managed to enhance their biodiversity, conservation and functional value for soil macrofauna? J Appl Ecol 45:269–278

Smith RG, McSwiney CP, Grandy AS, Suwanwaree P, Snider RM, Robertson GP (2008b) Diversity and abundance of earthworms across an agricultural land-use intensity gradient. Soil Tillage Res 100:83–88

Tian G, Olimah JA, Adeoye GO, Kang BT (2000) Regeneration of earthworm populations in a degraded soil by natural and planted fallows under humid tropical conditions. Soil Sci Soc Am J 64:222

Udawatta RP, Kremer RJ, Adamson BW, Anderson SH (2008) Variations in soil aggregate stability and enzyme activities in a temperate agroforestry practice. Appl Soil Ecol 39:153–160

van Capelle C, Schrader S, Brunotte J (2012) Tillage-induced changes in the functional diversity of soil biota—a review with a focus on German data. Eur J Soil Biol 50:165–181

Wotherspoon A, Thevathasan NV, Gordon AM, Voroney RP (2014) Carbon sequestration potential of five tree species in a 25-year-old temperate tree-based intercropping system in southern Ontario, Canada. Agrofor Syst 88:631–643

Zeithaml J, Pizl V, Sklenicka P (2009) Earthworm assemblages in an ecotone between forest and arable field and their relations with soil properties. Pesq Agrop Brasileira 44:922–927